

Divyanshu Raj

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SUMMARY

Computer Science Engineering student specializing in AI, Python automation, and Linux-based systems. Hands-on experience building AI-powered applications, scripting automation pipelines in Python, and working with embedded security architectures. Proficient in C, C++, Python, and Java with foundational knowledge of TCP/IP networking and wireless protocols. Seeking to contribute to AI engine development, network diagnostics, and Linux/embedded environments.

PROFESSIONAL EXPERIENCE

Pratinik Infotech

Machine Learning Intern

Feb 2026 – Apr 2026

Bengaluru, India

- Developed and automated ML data preprocessing pipelines in Python, reducing manual data preparation effort across multiple real-world datasets.
- Built, trained, and evaluated supervised ML models using Python libraries (scikit-learn, NumPy, Pandas); analyzed model performance metrics to guide iterative improvements.
- Implemented Python scripts to automate model testing and result logging workflows, improving reproducibility of experiments.
- Collaborated with senior engineers on model deployment workflows, gaining exposure to production-grade AI system integration.

PROJECTS

AI-Based Parental Control System

Python, Computer Vision, FastAPI, Android Security

- Designed and implemented an embedded-style intelligent security system in Python that automatically switches access modes using facial recognition, mimicking embedded system state-machine logic.
- Built a multi-layer authentication subsystem combining real-time face verification with OTP-based access control, implementing secure biometric storage with privacy-first design.
- Developed a FastAPI backend server running on Linux to handle authentication requests, activity logging, and uninstall protection – mirroring daemon/service design patterns on embedded Linux.
- Performed systematic threat analysis and loophole assessment to harden system security, improving authentication reliability and resistance to bypass attempts.

AI Code Review Assistant

Python, FastAPI, LLM APIs, REST

- Built a Python-based automation tool that interfaces with LLM APIs to analyze source code and deliver categorized, actionable feedback – automating a previously manual review process.
- Designed a RESTful FastAPI service on Linux to handle code ingestion, prompt routing, and structured response delivery, ensuring low-latency API communication.
- Developed an interactive dashboard for tracking review history and optimization recommendations, improving developer workflow visibility.
- Integrated context-aware LLM prompting to generate semantically relevant code insights across Python, Java, and C/C++ codebases.

WarrantyChain – Blockchain Warranty & Ownership Verification

Solidity, Ethereum, React.js, Node.js, Ethers.js

- Architected and deployed Solidity smart contracts on an Ethereum test network (Ganache) for warranty registration, verification, and ownership transfer with cryptographic auditability.
- Integrated MetaMask wallet-based authentication for secure, decentralized transaction approval, replacing paper-based warranty workflows.
- Built a React.js frontend and Node.js backend communicating with blockchain nodes via Ethers.js, establishing a full-stack decentralized application.

TECHNICAL SKILLS

Languages	C, C++, Python, Java, JavaScript
Networking	TCP/IP, Routing/Switching fundamentals, Wi-Fi protocols, REST APIs
Linux & Systems	Linux/Ubuntu, Bash scripting, process management, service/daemon workflows
AI & Machine Learning	Python ML pipelines, scikit-learn, Computer Vision, LLM APIs, OpenAI APIs
Embedded & Security	Embedded security design, biometric systems, authentication architecture
Web & Backend	FastAPI, Node.js, React.js, HTML, CSS
Databases	MySQL, MongoDB
Tools & Platforms	Git, GitHub, VS Code, Ganache, MetaMask, Ethers.js
Blockchain	Solidity, Ethereum, Smart Contracts

SOFT SKILLS

Team Collaboration • Leadership • Problem Solving • Adaptability • Time Management • Attention to Detail • Critical Thinking • Communication

EDUCATION

Acharya Institute of Technology
Bachelor of Engineering – Computer Science & Engineering / CGPA: 8.3/10

2023 – 2027
Bengaluru, India

Relevant Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems, Object-Oriented Programming, Software Engineering

CERTIFICATIONS

- Machine Learning Internship Certificate – Pratinik Infotech [\[View Certificate\]](#)
- Machine Learning and AI Foundations: Prediction, Causation and Statistical Inference – LinkedIn Learning
- Practical Cybersecurity for IT Professionals – LinkedIn Learning
- Introduction to Identity and Access Management (IAM) – LinkedIn Learning
- Build Your Cloud Computing Skills & Introduction to Cloud Computing for IT Professionals – LinkedIn Learning
- Networking and Administration Fundamentals – LinkedIn Learning
- Agile Project Management with Jira Cloud – LinkedIn Learning [\[View Certificate\]](#)

ACHIEVEMENTS

- Secured **Top 23 position** at **Hackaphasia Hackathon 2025** – competed against 200+ teams in a systems and AI-focused challenge.
- Participated in **AmBITion Hackathon 2025** – built a functional prototype under a 24-hour constraint demonstrating rapid problem-solving and systems design skills.
- Active participant in technical coding challenges, project-based competitions, and open-source learning programs.